Banning a Vanishing Margin: Brazil's Child Marriage Ban, Formal Marriage, and Informal Unions*

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Abstract

In 2019 Brazil eliminated the remaining pathways to civil marriage below age 16. We ask what public data can identify about the reform using the Civil Registry, PNADC, and SINASC. The locked Registry model estimates a 1.1% decline at age 15 (95% CI $[-14.9\%, +14.9\%]$), while omitting age-specific trends yields -37.8%; all five counterfactuals in a post-result rolling-origin protocol fail pre-law calibration. PNADC estimates a +0.399 percentage-point change in coresident-union prevalence (95% CI $[-0.003, +0.801]$). An exact-date SINASC protocol, frozen after the other results but before its own estimates, compares the age-16 discontinuity in 2016–2018 and 2022–2024. It estimates +0.342 points (95% CI $[-0.140, +0.824]$), but its MDE is 0.688 and its placebo intervals do not establish equivalence. Grouped SINASC fails its placebos. No design supports an unconditional causal estimate. The evidence identifies counterfactual limits when a national law targets an already declining formal margin.

JEL: J12, J13, K36, O15, I38. **Keywords:** child marriage; minimum-age laws; informal unions; civil registration; Brazil.

*Working draft; please do not cite or circulate. The primary Registry and PNADC specifications were SHA-256–locked on 2026-09-02 before any post-reform coefficient was computed. Later extensions are separately dated and hash-locked; the Registry trend sensitivity and exact-age SINASC design are explicitly post-result protocols frozen before their own estimates. [ACKNOWLEDGMENTS TO BE ADDED]

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1 Introduction

Minimum-age-of-marriage laws are a common policy response to child marriage, and eliminating child, early, and forced marriage is Target 5.3 of the Sustainable Development Goals [United Nations, Department of Economic and Social Affairs, 2026]. The evidence on whether these laws change behavior, however, is mixed at best: enforcement is weak in most of the developing world [Collin and Talbot, 2023], Mexico's state-level bans reduced *registered* child marriages while informal unions took their place [Bellés-Obrero and Lombardi, 2023], and an information experiment in Bangladesh found that publicizing a child-marriage reform could even backfire [Amirapu et al., 2026]. The core difficulty is that the law regulates a legal act—civil marriage—while the phenomenon of concern is a living arrangement that in much of the world requires no state involvement at all.

Brazil's Law 13,811 of March 2019 offers an unusually sharp version of this problem. The law closed the last legal pathways to civil marriage below age 16, which until then had been available in cases of pregnancy or to avoid the imposition or fulfillment of a criminal penalty. It did not touch marriages at ages 16–17, which remain lawful with parental consent. Two features make this reform a revealing case. First, treatment is defined by age alone and applies nationwide from a precise date, so an age-eligibility contrast is available in the universe of registration records. Second—and this is the substantive point—the margin the law banned was already close to extinction. In the pre-reform years 2013–2018, about 3,400 fifteen-year-olds married in the entire country, a rate of roughly 4 per 100,000; in the same period, household survey data put the share of 15-year-old girls living in a coresident union between 1.2% (a conservative definition) and 4.8% (a broad one). These quantities are not directly comparable—a registration rate is a flow and union prevalence is a stock—but together they show that the reform targeted a small formal margin alongside a larger observed stock of coresident unions. Vital records add a third measurement margin: among mothers younger than 16 giving birth in 2019, 0.7% declared themselves formally married, while 19% declared a *união estável*, the Brazilian informal-union status.

This paper asks what the available public data can identify about a national ban in that environment. The original Registry and PNADC specifications, treatment window, comparison groups, and inference procedures were locked—with cryptographic hashes— before any post-reform coefficient was computed, and every amendment is logged. The Registry design compares persons aged 15 with persons aged 17–19 within region and quarter, using the universe of registrations from 2013 to 2024. Because the reform is a single national event, regional cells are not independent policy experiments. We treat the Registry coefficient as conditional on a relative age-trend model and confront that model with national time-series forecasts, placebo dates, and alternative trends. We also use exact maternal birthdays in SINASC to estimate a post-minus-pre change in the age-16 discontinuity. That separate protocol was written after the Registry, PNADC, and grouped-age SINASC results were known but before any daily age-distance contrast was computed. It cannot replace the originally locked result merely because its estimate is more favorable.

Three results follow. First, the locked specification finds no detectable additional decline in marriage registrations at age 15 in the three fully treated quarters of 2019: the estimate is −1.1%, with a 95% confidence interval from −14.9% to +14.9%, and an implied 2.1 registrations averted (CI [−25.2, +33.8]). This is an imprecise null, not evidence of a zero: the design's minimum detectable effect at 80% power is a 19.3% decline. Conditional on the locked linear-trend specification, the confidence interval excludes declines larger than 14.9%, but it does not do so across admissible counterfactual trends. The same model without age-specific trends yields −37.8% ($p < 0.001$). A post-result forecast-validation protocol, frozen before its own estimates, ranks the seasonal-level no-trend analogue last on pre-2019 prediction. The best-ranked global-linear forecast gives +2.0% (95% forecast interval [−17.6%, +23.4%]), and an inverse-RMSE^2 ensemble gives +3.0%. Yet all five candidates fail the protocol's absolute calibration threshold. Their point estimates span [−33.7%, +12.8%], a specification envelope rather than a confidence or causal bound; the seasonal-level model's own interval, [−54.7%, −0.3%], includes a decline of 49%. The exercise therefore ranks extrapolations but cannot certify a counterfactual or rule out a large effect design-wide.

The closest benchmark in the literature, Mexico's staggered state bans, cut registered marriages of 16–17-year-old girls by 49 percent of the pre-reform mean [Bellés-Obrero and Lombardi, 2023]. That magnitude lies outside Brazil's locked-model interval, but the comparison is contextual rather than a cross-design hypothesis test. Mexico's significant effect arises at 16–17, where formal marriage was still common (roughly 1,700 registrations per 100,000 girl-years against Brazil's 4 per 100,000 at age 15),[1] while at 14–15—the margin comparable to ours—Bellés-Obrero and Lombardi [2023] likewise find no significant change. Read together, these age profiles are consistent with—but do not establish—the hypothesis that minimum-age laws have larger registration effects where formal marriage remains quantitatively important. We treat the failed calibration not as a robustness footnote but as a finding about how minimum-age laws can be mis-evaluated where the targeted margin is already declining.

Second, PNADC does not identify a reduction in conservative coresident-union prevalence. In quarterly household survey data, the point estimate among 15-year-olds is +0.399

[1] Derived from the published estimate: 0.695 fewer marriages per 1,000 girls per month over a 49 percent reduction implies a pre-reform mean of about 1.42 per 1,000 girl-months, or some 1,700 per 100,000 girl-years.

percentage points (95% CI $[-0.003, +0.801]$, $p = 0.052$) relative to ages 17–19. The sign is consistent with substitution from formal marriage toward informal cohabitation—in Mexico, the authors report that, among adolescent mothers, the decline in the formally married share was "completely counteracted" by an increase in the informal-union share [Bellés-Obrero and Lombardi, 2023]—but we are explicit about what this coefficient can and cannot say: the Registry measures a flow of formal events, the survey measures a stock of coresident arrangements, the two cannot be differenced, and equivalence testing shows the survey cannot rule out effects smaller than ± 0.5 p.p. The honest statement is that no reduction is detected, while the positive and window-sensitive estimate is consistent with relative informalization without establishing it.

Third, exact maternal birthdays in SINASC provide the paper's most local comparison. Among singleton live births within 90 days of the mother's sixteenth birthday, we compare the right-minus-left jump in recorded married status in 2016–2018 with the same jump in 2022–2024. The estimate is +0.342 percentage points (95% CI $[-0.140, +0.824]$; $p = 0.164$), in the direction predicted by the law, and all 13 frozen sensitivity estimates are positive ($[+0.148, +0.356]$). The interval nevertheless contains both zero and changes well above the ± 0.25-point policy margin; the MDE is 0.688 points. The delay-sensitive estimate is +0.323 points (95% CI $[-0.318, +0.964]$; Holm $p = 0.328$). Data integrity, density, and predetermined- composition gates pass. Counterfactual validation does not: neither the pre-law temporal placebo nor the age-15, 17, and 19 placebo-cutoff intervals establish equivalence at ± 0.25 points. The frozen protocol therefore classifies the counterfactual gate as qualified and the result as inconclusive; it does not authorize a causal core. The coarser grouped-age SINASC design is weaker still. Its trended and no-trend estimates span $[-23.9\%, +29.3\%]$, and its lead and placebo diagnostics reject the design.

The contribution is narrower than an effect estimate. To our knowledge, this is the first empirical evaluation centered on Brazil's 2019 ban,[2] We combine three independently auditable public sources but do not treat their different outcomes as replications. The resulting case is a limiting one: the law arrived after the formal flow had nearly disappeared, while the observable stock of coresident unions remained larger. The paper's methodological value is to show which comparisons survive contact with the data. The gap between −1.1%

[2] A systematic search on September 2, 2026—nineteen logged queries across economics (SSRN, NBER, IZA, CEPR, ANPEC), demography, public health (SciELO, PubMed), Brazilian legal scholarship, and forward citations of the closest international studies—found no causal evaluation of Law 13,811/2019. Existing empirical work is descriptive: Urquia et al. [2022] explicitly present their 2011–2018 perinatal series as a baseline against which the 2019 reform could later be measured, and a student report supervised by one of the authors [Faria, 2025] documents descriptively that the decline predates the law. We also verified that Brazil is not among the 18 countries in the multi-country ban panel of Le et al. [2025] (full text, IZA DP 17089).

and −37.8% exposes the Registry trend problem; the exact-age SINASC design improves the running variable and comparison locality but remains underpowered for the frozen margin and cannot affirm placebo equivalence. This complements evidence from settings with staggered reforms or larger formal margins [Bellés-Obrero and Lombardi, 2023, Collin and Talbot, 2023] without treating a national before–after contrast as if it contained untreated policy variation.

The rest of the paper proceeds as follows. Section 2 reconstructs the legal change from primary sources. Section 3 describes the three data sources and the measurement decisions, which in this setting carry identification weight. Section 4 states the estimands, identifying assumptions, and evidentiary hierarchy. Section 5 reports Registry results; Section 6 turns to coresident unions and status at birth; Section 7 collects robustness and threats. Section 8 discusses what the results imply for minimum-age laws and their monitoring.

2 Institutional background

Under the Civil Code of 2002, the marriageable age in Brazil is 18, or 16–17 with the authorization of both parents or legal representatives (art. 1.517) [Brasil, 2002]. Before 2019, art. 1.520 allowed marriage *below* 16 in two exceptional situations: to avoid the imposition or fulfillment of a criminal penalty (a clause tied to the pre-2005 treatment of sexual offenses) and in case of pregnancy. Law 13,811, published and in force on March 13, 2019, rewrote art. 1.520 to eliminate both exceptions: since then, no civil marriage below 16 can be lawfully celebrated in Brazil [Brasil, 2019].

Three institutional facts matter for the design. First, the law changed eligibility for a formal act performed before a civil registrar; it created no new enforcement apparatus directed at informal unions, and *união estável* remains a constitutionally recognized family form with no minimum-age registration requirement. Second, the treated population is precisely delimited by age at the date of celebration: persons under 16. Ages 16–17 are legally untouched, though behaviorally they may absorb postponed marriages, which is why our primary comparison group is ages 17–19 and contrasts involving 16–17 are interpreted as potentially contaminated by delay. Third, the reform was national and immediate, with no phase-in, no state discretion, and no grandfathering of pending authorizations that we could verify in primary sources; the identification problem is therefore the single-event problem, not the staggered-adoption problem, and we treat it as such throughout.

The registry data also show what the law was regulating. Marriages with at least one spouse below 16 fell from 1,093 in 2013 to 512 in 2018—before any reform—and to 419 in 2019 and 193 in 2024. The decline that matters for Brazilian adolescents was well under way for reasons the law did not cause, which is precisely why an evaluation that does not model it will misattribute it.

3 Data and measurement

Our three sources measure different objects. We keep them separate by construction and never subtract a stock coefficient from a flow coefficient. Every estimate and exhibit in this manuscript uses observed official microdata or published aggregates; no simulated observation enters the empirical evidence.

3.1 Civil marriage registrations (flow of formal events)

The Civil Registry statistics (IBGE, SIDRA table 4406) record every civil marriage by single year of age of each spouse, sex, month, and place of *registration*, from 2013 to 2024. We reconstructed the official aggregates exactly: all 3,780 local age–sex cells and every verifiable local total match the published figures. Age refers to age at registration in completed years; the month is the month of registration, not celebration; and the geography is the registry office, not the couple's residence. Each of these measurement facts constrains the design: completed years rule out a conventional regression discontinuity, registration lags blur the treatment date (March 2019 is excluded; the post window opens in April/2019Q2), and office-versus-residence geography caps the defensible spatial resolution at the region level, per a precision rule fixed before estimation.

3.2 PNADC (stock of coresident unions)

The continuous national household survey (PNADC) provides quarterly cross-sections of 2.43 million adolescents aged 14–19 over 2012–2024 (48 quarters; calibrated weights; design-based variance using strata and PSUs). Our conservative union indicator equals one when the adolescent is the spouse/partner of the household head or is the head with a coresident partner; a broader indicator adds plausible nested pairings and is used only as robustness with an explicit ambiguity flag. The survey follows dwellings, not individuals; an adolescent who leaves home to form a union elsewhere may exit the sampled dwelling, so the indicator undercounts unions in a way that can correlate with the outcome. We therefore treat PNADC as measuring the stock of *coresident, household-grid-visible* unions, and we report equivalence tests and minimum detectable effects rather than interpreting non-significance as absence of effect.

3.3 SINASC (conjugal status at birth)

The national live-birth information system (SINASC, Ministry of Health) records every live birth with the mother's age—and, since the 2011 form, her exact date of birth—her declared conjugal status (single; married; widowed; separated/divorced; *união estável*; unknown), municipality of *residence*, and the child's date of birth, along with education, race, parity, and prenatal-care covariates. We use the official open microdata, 2013–2024 (2.4–2.9 million births per year; SHA-256-verified acquisition; layouts harmonized across the 2015/2016 file-generation break; audit in the replication package). Cross-validation against published SVSA totals confirmed the open files to the record for 2014 but revealed the 2015 export to be materially incomplete; 2015 is excluded from estimation under a logged data-error amendment (Section 6). Two caveats govern interpretation. Conjugal status is declared at the birth, without registry verification: it measures the stock of statuses among women giving birth, not a marriage flow, and post-law declarations could re-label rather than reflect behavior. And any SINASC estimand conditions on a birth having occurred, so it is a joint outcome of fertility and formalization; we present it as such. In exchange, SINASC observes 55–70 thousand births to mothers under 16 per year—against 194 treated-cell marriage registrations in 2019Q2–Q4—with residence geography and daily dates, which no other source offers.

The exact-age design uses only records with valid birth dates for mother and child, valid conjugal status, valid municipality of residence, and singleton gestation. The running variable is the integer number of calendar days between childbirth and the mother's sixteenth birthday; births on the birthday enter the eligible side. The primary sample contains 126,138 observed births within 90 days of the cutoff and 1,342 recorded- married outcomes, across 5,183 municipalities and 2,192 exact birth dates. The pre-law window is 2016–2018 and the mature post-law window is 2022–2024. We exclude 2019 as a transition year and 2020–2021 as pandemic years rather than ask fixed effects to absorb them. For all six primary years, raw-file row counts match independent official annual totals exactly; at least 98.86% of records have valid exact dates, and completed age agrees with calendar age for at least 99.98% of valid-date records.

3.4 Denominators and comparability

Population denominators by age, sex, region, and quarter come from PNADC with a cell-quality rule (minimum unweighted counts and maximum coefficients of variation) frozen before estimation; state-level cells failed the rule and region was selected as the primary geography. The under-15 registry category is reported as counts and shares only—no rate with a fabricated 0–14 population is ever computed.

4 Empirical strategy and identification limits

4.1 Identification hierarchy

No source supplies an untreated jurisdiction: Law 13,811 changed the legal regime nationwide on one date. Identification must therefore come from an age contrast, a model of its counterfactual evolution, or both. Table 1 states the variation and indispensable assumption behind every empirical object. The Registry specification remains primary because it was locked first, not because later diagnostics validated its counterfactual. PNADC and the two SINASC designs measure different populations and outcomes; agreement among their coefficients would not turn them into replications of one causal estimand.

Table 1: Identification hierarchy and maximum defensible interpretation.

Evidence	Source of variation and estimand	Indispensable counterfactual assumption	Status
Civil Registry	Age 15 versus ages 17–19 in 2019Q2–Q4; ratio and level change in registration flows	The age-15 gap would follow the specified relative trend, with no coincident age-specific national shock or differential registration lag	Model-conditional; all five rolling-origin candidates fail calibration
Exact-age SINASC	Post-minus-pre change in the right-minus-left age-16 jump among singleton live births within 90 days	Smooth potential recorded status at the birthday and a stable non-law age-16 jump across 2016–2018 and 2022–2024; no differential birth selection	Local design; G0/G1 pass, G2 qualified, G3 inconclusive
PNADC	Age-15 versus ages 17–19 change in coresident-union prevalence	Correct relative prevalence trend and stable survey observation of adolescents who form unions	Suggestive stock contrast; not evidence of substitution
Grouped-age SINASC	Mothers 10–15 versus 17–19 change in declared status at birth	Correct group-specific trend and stable composition of mothers giving birth	Placebos reject; descriptive only

Notes: Registry outcomes are formal-event flows; PNADC is a coresident-union stock; SINASC is recorded status conditional on a live birth. None of the four rows supports an unconditional causal estimate in the present data.

4.2 Registry estimand and source of variation

Treatment is age eligibility: a 15-year-old could, in principle, marry under the pregnancy exception before March 13, 2019 and cannot since; a 17-year-old's legal regime is unchanged.

Let M_{rat} denote registrations in region r, age a, and quarter t, and let P_{rat} be the corresponding population. The locked PPML model is

$$\mathbb{E}[M_{rat} \mid P_{rat}, r, a, t] = P_{rat} \exp\big(\alpha_{ra} + \lambda_{rt} + \gamma_{aq(t)} + \delta_a t + \beta D_{at}\big)\,, \tag{1}$$

where D_{at} equals one for age 15 in 2019Q2–Q4, $q(t)$ is calendar quarter, α_{ra} are region-by-age fixed effects, λ_{rt} are region-by-period fixed effects, $\gamma_{aq(t)}$ are age-specific seasonal effects, and $\delta_a t$ are age-specific linear trends. The primary estimand is $100[\exp(\beta) - 1]$, the short-run percentage change in age-15 registrations relative to ages 17–19, together with the corresponding change in registrations per 100,000 residents and in event counts.

The identifying variation is the age-15 deviation in three post-law quarters after the model extrapolates relative age trends from 24 pre-law quarters. The error contains any national shock in those quarters that changes age-15 registration differently from ages 17–19, including an age-specific continuation of secular decline. Region-by-period fixed effects cannot absorb such a shock. The controls are legally untreated but need not be behaviorally unaffected: postponement into those ages would also violate the comparison. We exclude age 16 from the primary controls because it is the most likely recapture age. We omit 2019Q1, which is partly treated, and end the primary window in 2019Q4; 2020–2021 is pandemic-contaminated. Alternative controls, no-trend and linear-rate variants, monthly frequency, and pre-windows starting in 2014 and 2015 were frozen before estimation.

4.3 Exact-age SINASC difference in discontinuities

The daily design asks a narrower question: did the law increase the age-16 jump in recorded married status among adolescent mothers giving birth? For mother i, let x_i be the integer number of calendar days from childbirth to her sixteenth birthday, $A_i = 1[x_i \geq 0]$, and Q_i indicate the mature post-law years 2022–2024 rather than 2016–2018. Within $|x_i| < 90$, the triangular-weighted local-linear model is

$$\begin{aligned} Y_i = \alpha + \theta A_i + \tau(A_i Q_i) + b_1 x_i + b_2(A_i x_i) \\ + b_3(Q_i x_i) + b_4(A_i Q_i x_i) + \eta_{y(i)} + \mu_{m(i)} + \varepsilon_i, \end{aligned} \tag{2}$$

where Y_i equals 100 if SINASC records the mother as married and zero for another valid status, $\eta_{y(i)}$ are birth-year fixed effects, and $\mu_{m(i)}$ are child- calendar-month fixed effects. Standard errors are two-way clustered by municipality of residence and the child's exact birth date. The coefficient τ is the post-minus-pre change in the right-minus-left limit at age 16, in percentage points.

This is a difference in discontinuities, not a conventional sharp RD of marriage. Age at childbirth is not age at marriage, actual marriage is unobserved, and the age-16 legal threshold exists in both eras. A causal reading of τ requires the non-law component of the age-16 jump to remain stable across the two windows, in addition to local smoothness within each window. It also requires no reform-induced discontinuity in who gives birth and no post-pandemic shock that changes recorded status precisely at age 16. Density, predetermined composition, status missingness, a pre-law temporal placebo, and placebo birthdays at 15, 17, and 19 were therefore binding gates. A delay-sensitive secondary estimand averages the fitted excess over the first 90 days above the birthday; it cannot replace the cutoff result or repair a failed gate.

4.4 Inference for one national reform

The Registry cells share one treatment date and do not constitute independent policy experiments. Primary inference clusters on time periods (27 clusters, only 3 post). We compare it with an aggregate Brazil series using HAC standard errors, a temporal block bootstrap of the pre-period forecast, pre-law placebo dates, and randomization inference over feasible dates. These procedures characterize sampling and forecasting uncertainty; none makes a misspecified trend into a counterfactual. A two-way cluster variant produced an implausibly small standard error after a positive-semidefinite adjustment with five regions, so it receives no evidentiary weight. The daily SINASC clustering follows its different source of local variation and does not inflate the number of national reforms.

4.5 Specification locks and result chronology

The Registry and PNADC estimands, samples, windows, comparison groups, models, and inference were frozen and hash-stamped before any post-reform coefficient was computed (2026-09-02T00:56; SHA-256 hashes in the replication package). Logged amendments concern computational identification, not coefficient sign or significance. The grouped-age SINASC extension was separately frozen on September 2 after its data audit and before its post-reform contrasts. The daily SINASC protocol was frozen on September 4 after all earlier paper results were known but before any daily age-distance outcome contrast, density result, or placebo-cutoff estimate was viewed. Its post-result status, expected signs, gates, and stopping rule remain visible because a later lock is not ex ante with respect to the paper.

5 Registry results

5.1 Raw series and the vanishing margin

Figure 1 plots registration rates by single age. The age-15 series is one to two orders of magnitude below ages 17–19 throughout and declines over the whole pre-period. In the 2013–2018 window, 3,394 fifteen-year-olds married nationwide (4.19 per 100,000 per year), against 94,107 sixteen-year-olds and 130,000 seventeen-year-olds. The under-15 category shrinks from 364 marriages in 2013 to 246 in 2016 and 85 in 2024.

5.2 The locked estimate

The primary PPML contrast for 2019Q2–Q4 yields a rate ratio of 0.989: -1.1%, 95% CI $[-14.9\%, +14.9\%]$, $p = 0.888$ (period clusters). In levels this is -0.022 per 100,000, or 2.1 registrations averted against a counterfactual, with CI $[-25.2, +33.8]$. The aggregate HAC series gives $+2.0\%$ ($p = 0.785$) and the temporal bootstrap forecast $+2.0\%$ ($p = 0.857$). Figure 2 shows the event study: no post-reform break stands out from the pre-period variation.

5.3 Trend dependence as a result

Removing the age-specific linear trends flips the estimate to -37.8% ($p < 0.001$). Monthly frequency with trends gives -2.4% ($p = 0.820$). Starting the pre-window in 2014 or 2015 moves the locked estimate to $+8.2\%$ and $+11.1\%$. The joint test of 24 leads does not reject ($p = 0.161$), but no simultaneous band stays within $\pm 10\%$, and one placebo date (2015Q2) is significant ($p = 0.002$): parallel evolution cannot be certified, it can only be modeled, and the conclusion depends materially on how. This is the central empirical lesson of the paper: in a collapsing series, the counterfactual trend *is* the estimate.

We therefore discipline the trend choice with a forecast-validation exercise, frozen as a post-result protocol before any of its sensitivity estimates were computed (five candidate counterfactuals for the national age-15 versus 17–19 log-rate gap; six overlapping three-quarter rolling windows over 2017Q1–2018Q2, fit strictly before each window; calibration tiers, ensemble weights, and bootstrap fixed in advance; Figure 3 and Appendix Table 6). The exercise disciplines in both directions. It discriminates: the seasonal-level candidate—the forecast analogue of the PPML no-trend specification—is decisively the worst forecaster (rolling RMSE 0.391 against 0.131 for the best; zero of six windows within a rate-ratio factor of 1.20). The four lower-RMSE trending models put the 2019Q2–Q4 deviation between $+1.1\%$ and $+12.8\%$; their model-specific p-values are at least 0.41, and the inverse-RMSE^2

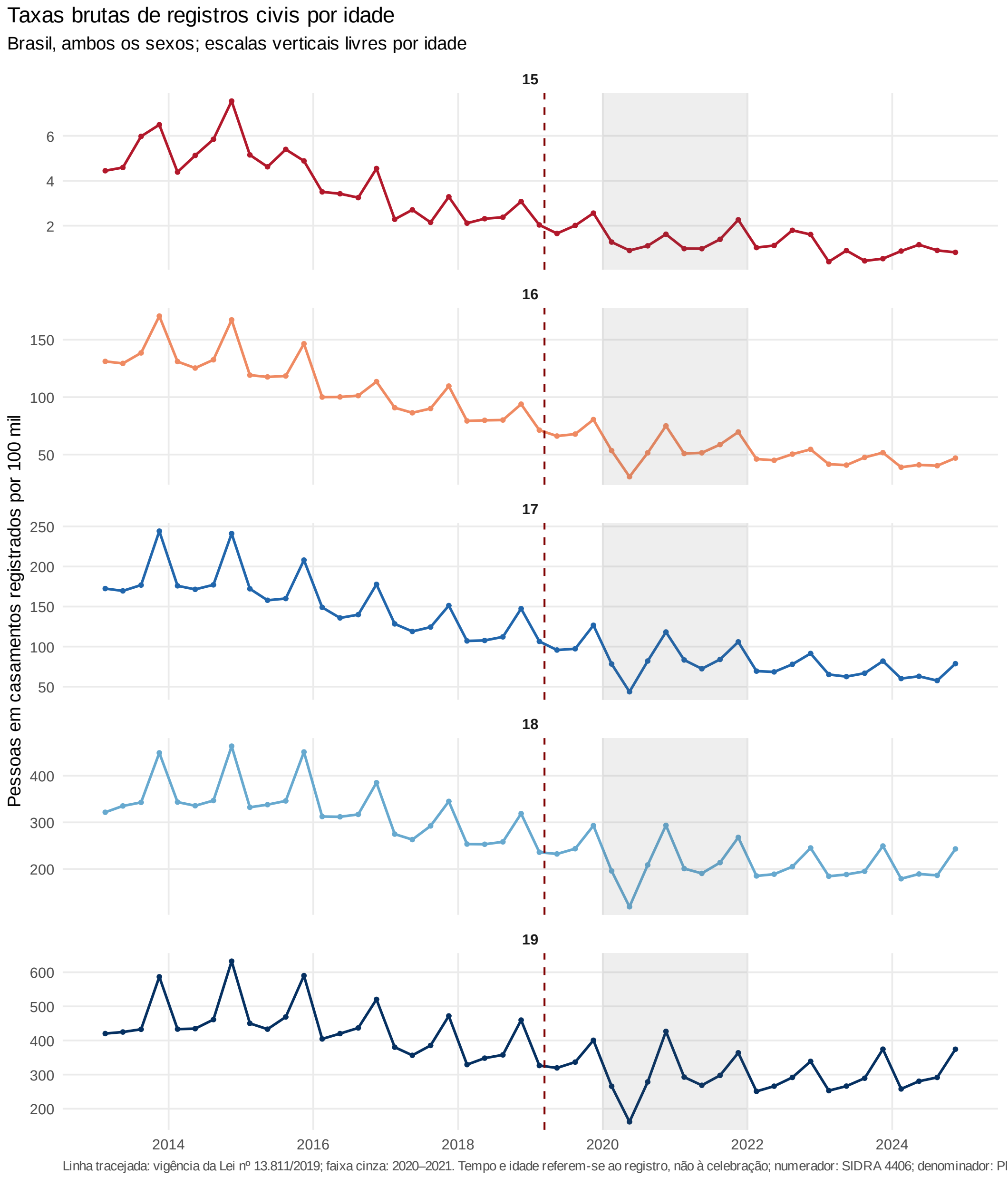

Figure 1: Civil marriage registrations per 100,000 residents, by single age at registration. Vertical line: Law 13,811 (March 2019). Source: SIDRA 4406 and PNADC denominators; replication package.

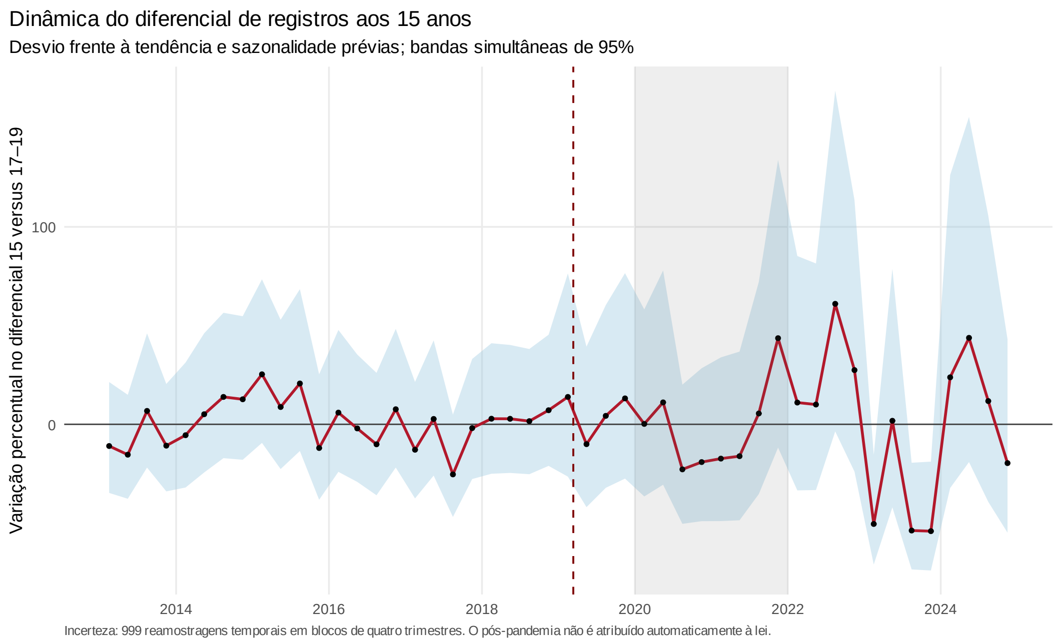

Figure 2: Event study, age 15 versus 17–19, locked specification. 2019Q1 omitted; post window 2019Q2–Q4. Source: replication package.

ensemble gives +3.0%. And it humbles: no candidate meets the absolute calibration bar fixed in the protocol (the best model's worst window still misses by a factor of 1.26), so the calibration gate *fails*. No cross-model range earns the name of confidence set—we report the point-estimate range $[-33.7\%, +12.8\%]$ as a specification envelope only. Moreover, the seasonal-level forecast interval $[-54.7\%, -0.3\%]$ includes the 49% Mexico benchmark. The same standard we apply to SINASC below therefore binds here symmetrically: the Registry counterfactual cannot be certified either. The difference between the two sources is one of degree, dictated by their diagnostics—four trending Registry models cluster near zero, while SINASC's placebo battery rejects its design outright—not one of asymmetric treatment.

5.4 Power, and the absence of recapture

The design's 80%-power MDE is a 19.3% decline (power is 28% against a 10% decline and 83% against 20%). These numbers acquire meaning against the literature's benchmark: the 49 percent drop in registered marriages that Mexico's bans produced at their bound margin [Bellés-Obrero and Lombardi, 2023] lies outside the locked-model confidence interval and is far beyond the region where that model has high power. But the MDE describes sampling precision conditional on the locked trend; it does not repair counterfactual identification. Because every forecast candidate fails calibration and the seasonal-level interval includes a 49% decline, the full exercise cannot rule out that benchmark design-wide. What the locked

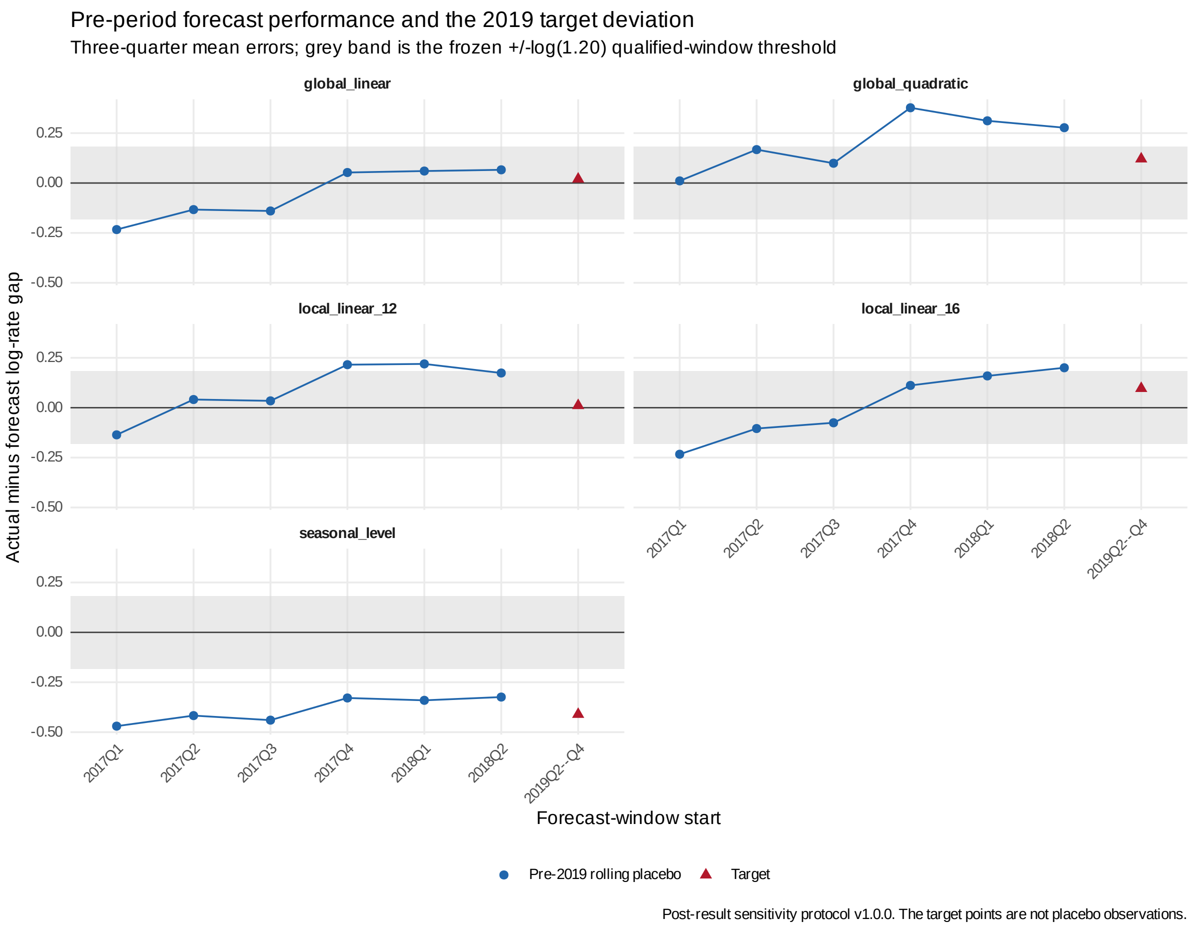

Figure 3: Rolling-origin validation of five counterfactual trend models. Blue circles are actual-minus-forecast mean log-rate gaps in six pre-2019 three-quarter windows; red triangles are the 2019Q2–Q4 target deviations. The gray band is the frozen $\pm\log(1.20)$ qualified-window threshold. Model fitting never uses the forecast window. Source: post-result sensitivity protocol and replication package.

model cannot detect are effects below roughly one-fifth of the baseline, which at Brazil's age-15 rate of 4 per 100,000 correspond to a handful of marriages per year. Effects at ages 16–19 are +0.6%, +3.3%, +0.8%, and −2.4%, none surviving familywise correction; the aggregate "recapture" ratio (excess at 16–17 over deficit at 15) is undefined in half of the bootstrap draws and spans $[-114, +1{,}107]$ where defined. There is no reliable evidence of postponement toward legal ages, and with 2.1 point-estimated averted events there is little to recapture. By sex, the female estimate is −9.7% (Holm $p = 0.125$); the male +155.8% is driven by 36 zero cells on a rare base and is not interpretable.

It is worth stating the stakes of this imprecision in persons rather than percentages. The locked levels estimate is 2.1 registrations averted over 2019Q2–Q4, with a confidence interval from −25 to +34 events—roughly −34 to +45 marriages on an annualized basis, against 419 under-16 marriages registered in all of 2019. Effects that would matter for hundreds of Brazilian adolescents over a decade therefore live comfortably inside the locked-model interval. Conditional on that trend model, the data reject a short-run transformation on the scale documented elsewhere; across the failed counterfactual candidates they do not. Both low event counts and counterfactual uncertainty bind: a national ban aimed at a few hundred registrations per year cannot produce effects measured in thousands, while the observed secular decline prevents us from attributing even the remaining movements to the law.

5.5 The longer run, and why we do not claim it

The diagnostic event study continues past the primary window, and its 2022–2024 coefficients are large: rate ratios between roughly 0.21 and 0.32 in several quarters, against 17–19-year-old controls. Descriptively, under-16 marriages fell from 419 in 2019 to 193 in 2024. We deliberately attribute none of this to the law. The intervening pandemic moved registrations, unions, and births in age-specific ways no fixed effect absorbs; the long-run coefficients carry five-region diagnostic inference that Amendment 1 already disqualified for formal conclusions; and the locked windows designate 2020–2021 as contaminated and 2022–2024 as descriptive. A reader who wants to see the long run finds it in the replication package, plotted and tabulated; a reader who wants a causal long-run estimate needs a design that separates the reform from the pandemic, which these data do not provide.

Table 2: Locked primary age-based PPML estimate, 2019Q2–Q4.

Model	Rate ratio (95% CI)	Change (95% CI)	Level effect per 100,000	Events averted (95% CI)	p
Locked PPML	0.989 [0.851, 1.149]	−1.1% [−14.9, 14.9]	−0.022	2.1 [−25.2, 33.8]	0.888

Notes: Region × age × quarter PPML with population offset; ages 17–19 are controls. Inference clusters by period (27 periods; five regions). The confidence interval for events averted reverses the sign of the corresponding event-effect interval.

6 Unions and formalization at birth

6.1 Coresident unions (PNADC)

The point estimate for the conservative union indicator among 15-year-olds is positive after the reform. The design-based cell estimate is +0.399 p.p. (95% CI $[-0.003, +0.801]$, $p = 0.052$); the aggregate HAC series gives +0.235 p.p.; microdata LPMs with household-period clustering or full survey-design variance give +0.344 p.p. (CIs $[-0.119, +0.807]$ and $[-0.025, +0.712]$); the broad indicator gives +0.597 p.p. ($p = 0.102$). The MDE is 0.575 p.p. and equivalence at ±0.50 p.p. fails (TOST $p = 0.311$). Estimates shrink as the pre-window shortens (0.399, 0.329, 0.113 p.p. for windows starting 2013, 2014, 2015). We conclude: no evidence of a reduction in conservative coresident-union prevalence; a positive, window-sensitive signal consistent with relative informalization that the survey cannot establish at conventional strength.

6.2 Exact-age eligibility at childbirth (SINASC)

Figure 5 plots observed weekly shares and local fits around the mother's sixteenth birthday. Before any married-status contrast was estimated, the frozen data gate reconciled all six primary-year files exactly to independent official annual totals. The density ratio at the cutoff is 0.994 before the law and 0.963 in the mature post period (robust $p = 0.848$ and $p = 0.395$). None of ten predetermined race or maternal-birthplace continuity tests approaches the joint significance-and-magnitude failure rule, and the change in missing-status discontinuity is +0.262 points ($p = 0.317$). Counts over 30 equal days differ across sides by about 6.2% in both eras, which triggers the frozen heaping diagnostic; one-, three-, and seven-day donut estimates therefore remain visible, though they cannot replace the primary sample. These checks support local comparability at the birthday but cannot validate change across eras.

The primary post-minus-pre change in the age-16 jump is +0.342 percentage points (SE 0.246; 95% CI $[-0.140, +0.824]$; $p = 0.164$). It equals 37.6% of the triangular-weighted pre-law below-cutoff married share of 0.909%. The interval contains zero and changes more than

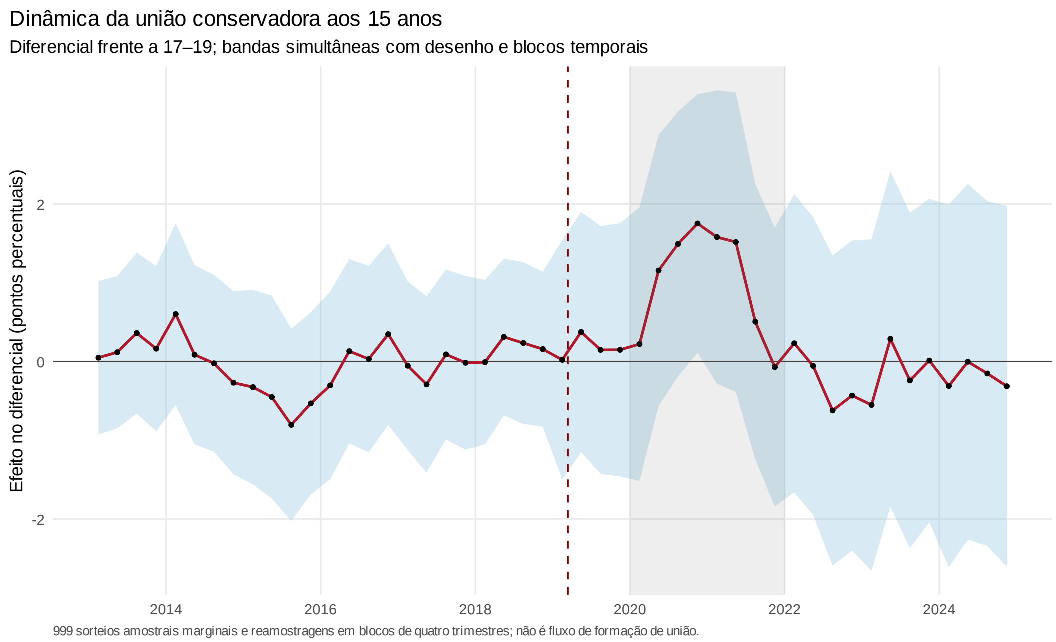

Figure 4: Coresident-union prevalence at age 15 (conservative definition), event study relative to ages 17–19. Source: replication package.

three times the frozen 0.25-point relevance threshold; the 80%-power MDE is 0.688 points (Appendix Table 3). All 13 frozen specifications identify and remain positive, spanning $[+0.148, +0.356]$ points, but none changes the primary inference (Appendix Figure 7). The delay-sensitive 90-day profile is +0.323 points (95% CI $[-0.318, +0.964]$; Holm $p = 0.328$), and a joint test of the immediate jump and slope kink gives $p = 0.379$. Separate bias-corrected local-polynomial fits imply a similar +0.325-point post-minus-pre difference, but the separate fits do not retain the cross-era covariance, so that number receives no independent inferential weight.

The binding limitation is counterfactual precision. A pre-law pseudo-reform gives −0.280 points with a 90% interval of $[-0.725, +0.165]$. Placebo birthdays at ages 15, 17, and 19 give −0.027, +0.054, and −0.064 points; none rejects after Holm adjustment, but none has a 90% interval contained in the frozen ±0.25-point equivalence band. Annual age-16 jumps also vary: +0.951 points in 2022, +0.113 in 2023, and +0.404 in 2024 (Appendix Figure 6). Under the rule written before these estimates, non-rejection at false cutoffs is not affirmative validation. G2 is therefore *qualified*, G3 is *inconclusive*, and the exact-age result is a positive but imprecise local signal, not an estimated causal effect of the law. The recorded *união-estável* and any-union changes are likewise imprecise (−0.628 and −0.286 points; Holm $p = 1.00$ for both) and do not identify relabeling or behavioral substitution.

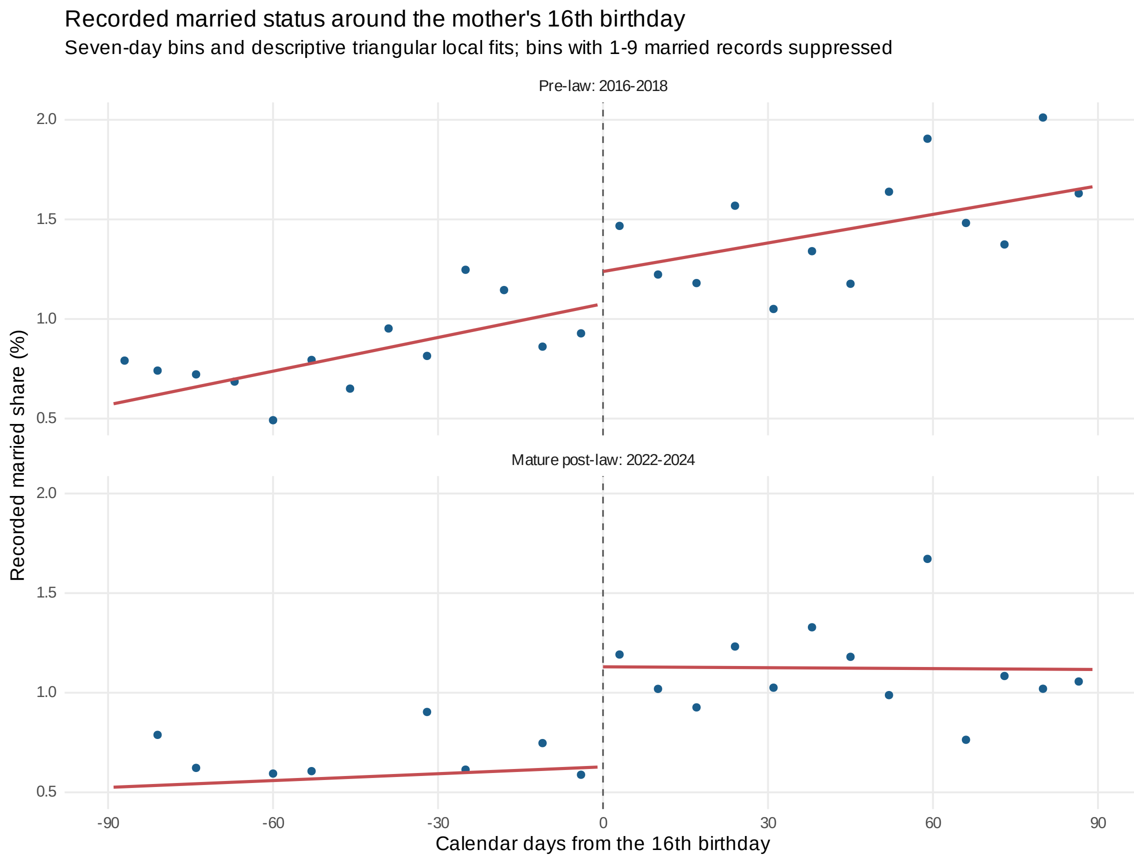

Figure 5: Recorded married status around the mother's sixteenth birthday. Points are observed seven-day-bin shares for singleton live births in 2016–2018 and 2022–2024; bins with 1–9 married records are suppressed. Lines are descriptive triangular local- linear fits within 90 days. The primary estimate uses individual records, birth-year and calendar-month fixed effects, and two-way clustering by municipality and exact birth date. It is the post-minus-pre change in the cutoff jump, not either displayed jump alone. Source: official SINASC microdata and replication package.

6.3 Grouped-age status contrast (SINASC)

Among the roughly 55–80 thousand mothers under 16 giving birth each year, the formally married share was already small and falling before the law: 1.72% in 2013 (1,345 of 78,062), 0.88% in 2016 (593 of 67,419), and 0.69% in 2019 (374 of 54,166), against informal-union shares of 27.9%, 23.2%, and 19.5%. Mothers aged 16–17 in 2019: 3.9% married, 23.3% in *união estável*. Monthly counts of married under-16 mothers in 2019 (19–47 per month) show no visible April collapse, consistent with status-at-birth being a slowly decaying stock among parturients rather than a marriage flow. Validation against the official SVSA epidemiological bulletin uncovered a material defect in the open microdata: the 2014 file matches the published total to the record (2,979,259), but the 2015 export is short by an implied 231,143 records (7.7%), concentrated in October–December; calendar year 2015 is therefore excluded from every estimate below, under a data-error amendment logged before re-estimation.

The pre-registered contrast (extension lock v1.0.0: mothers 10–15 versus 17–19, region-of-residence by month PPML on counts of married births with valid-status births as exposure, the registry's fixed-effect and trend structure, period clustering) returns +29.3% (95% CI $[+11.3\%, +50.1\%]$, $p < 0.001$; 270 married births in the treated post cells; Appendix Tables 4–5). We recorded the expected sign as negative-or-null before estimation. The positive estimate runs against the direct eligibility mechanism, while conditioning on birth leaves room for composition; the frozen diagnostics decide whether the contrast is credible, and they reject it. The monthly event study (Figure 9) shows the treated married share falling steeply through 2016 and then running flat at -30 to -37% of its pre-2017 level, straight across the reform date; the joint test of 24 leads has $p \approx 0$; and placebo bans dated April 2016, 2017, and 2018 return -5.9% (n.s.), +19.7% ($p = 0.004$), and +36.8% ($p < 0.001$)—a monotone-in-time pattern that is the signature of a linear trend extrapolating a convex, floor-approaching decay, not of treatment. The no-trend variant returns -23.9%; every trended variant (alternative controls, treated age 15 only, exact maternal age from dates of birth, quarterly aggregation, missing-as-unmarried) sits between +22% and +33%—except the variant whose pre-window starts in 2016, after the flattening, which returns +1.5% ($p = 0.86$), exactly as the artifact reading predicts. We therefore report $[-23.9\%, +29.3\%]$ as a specification range, not a causal confidence interval. The *união-estável* and any-union contrasts (+3.7% and +5.1%) inherit the same defect and receive no causal reading; their point estimates sit inside the pre-set equivalence margin of a 0.85–1.176 rate ratio (TOST $p < 0.001$), while the married contrast does not ($p = 0.89$), but under a rejected design equivalence is a description of the estimates, not of the reform. Two frozen diagnostics survive: the fertility-composition contrast—births to mothers aged 15 per 100,000 girls aged 15—is +1.6% (95% CI $[-1.3\%, +4.6\%]$; $p = 0.284$), and unknown-status reporting shows no detectable differential change ($p = 0.577$). What SINASC establishes, then, is descriptive and disciplinary at once: the formal margin at birth was small and had stopped falling *before* the ban, and even a universe-sized administrative dataset cannot rescue a short-run contrast whose counterfactual trend is the conclusion. The monthly event study appears in Appendix Figure 9.

7 Robustness and threats to identification

Several threats are bounded or made visible, not eliminated: registration lag (omit March/Q1; monthly variant), pandemic contamination (the Registry primary window ends in 2019Q4), geographic mismatch (region-level Registry analysis; municipal contrasts blocked), denominator uncertainty (repeated draws; SD an order below the temporal SE), and multiplicity (Holm within families). A pre-exposure triple-difference—regions with higher pre-law shares of under-16 marriages—shows a positive but imprecise gradient (+9.9% per SD, $p = 0.332$) and a -0.618 holdout correlation between exposure and 2018 changes, indicating mean reversion; it does not identify the national effect and is reported as descriptive.

The exact-age SINASC comparison removes the Registry's completed-age and office-location problems, but it changes the estimand. It conditions on a singleton live birth, measures status at childbirth rather than marriage incidence, and compares 2016–2018 with a post-pandemic window. The observed density and predetermined covariates are continuous under the frozen failure rules, yet unobserved birth selection may still change across the birthday and era. Marriage processing also creates an unknown lag between legal eligibility and recorded status: the delay-sensitive profile addresses a 90-day linear pattern only. Most important, wide temporal and placebo-cutoff intervals fail to affirm the required counterfactual stability. These limits are why a visually larger post-law jump does not become an RD effect.

Two timing threats deserve their own accounting. Anticipation in the rush-to-marry direction—couples accelerating a marriage before the ban—is not visible in the data: the diagnostic event-study coefficients for 2018Q3 and 2018Q4 are -0.7% and -1.9% (both $p > 0.80$), and the partially treated transition quarter 2019Q1, which the locked estimand omits, shows a -25.2% dip ($p = 0.02$, five-region diagnostic inference only), consistent with the law biting within its own quarter of entry rather than being front-run. The opposite threat—*de facto* tightening before the *de jure* change, with judges increasingly refusing the pregnancy-exception authorizations that were the only gateway below 16—cannot be tested with our data: if courts had largely closed the gate before March 2019, treatment is misdated, part of the pre-period path already reflects enforcement, and our short-run contrast understates the total effect of the legal regime. We flag this as an unverifiable threat. The public DataJud API exposes the relevant process class, but only for public proceedings, while Brazilian civil procedure places cases concerning marriage under judicial secrecy; moreover, the mandated historical load does not guarantee cases closed before 2020 [Conselho Nacional de Justiça, 2026, Brasil, 2015, Conselho Nacional de Justiça, 2020]. Our aggregate probe of all 27 state-court endpoints is not a usable workaround: 94.1% of 3,574 public class-143 records had filing dates after the September 2, 2026 snapshot, and only 179 had plausible dates in 2013–2024. Secrecy-inclusive, age-specific records of requests and decisions would directly test the de facto-timing threat, but no such research-ready public file is available. A higher-return redesign is feasible in principle: the restricted IBGE Registry instrument collects exact celebration and birth dates and spouse residence, and IBGE identifies its controlled-access room as the route for external researchers [Instituto Brasileiro de Geografia e Estatística, 2022, 2018]. Year-by-year retention and project approval remain outstanding. The remaining named threats are age measured in completed years; nationally synchronized age-specific shocks, which no fixed effect absorbs; and the selection inherent in any status-at-birth outcome.

8 Discussion and conclusion

Brazil banned civil marriage below 16 after registrations at that age had become rare, while coresident unions remained more prevalent on a conceptually different stock measure. The locked Registry model estimates -1.1%, but every rolling-origin counterfactual fails calibration. The exact-age SINASC comparison improves the running variable and local comparison: its $+0.342$-point estimate has the sign predicted by the law and stays positive across the frozen sensitivity set. Yet its interval $[-0.140, +0.824]$ is too wide to distinguish zero from a policy-relevant change, and the placebo intervals do not establish the stability required for a difference in discontinuities. PNADC does not identify a reduction in coresident-union prevalence, and the grouped-age SINASC design fails its placebo battery. The common conclusion is about the evidence, not the law's effect: current public data establish neither an additional decline caused by the reform nor the absence of one.

Two lessons travel beyond Brazil. First, a minimum-age law changes eligibility for a formal act. Its evaluation must keep civil-registration flows, coresident-union stocks, and declared status among mothers giving birth separate; no algebra turns one into the other. Second, a nationwide reform supplies only one policy date. Region-by-time cells do not create untreated reforms, and exact birthdays do not remove the need to identify how an age discontinuity would have evolved. In this application, specification locks prevent two tempting substitutions: promoting the -37.8% no-trend Registry estimate, or promoting the positive daily SINASC estimate after seeing it.

A descriptive companion exercise in the replication package puts the Brazilian case in its place among reforming countries—and the place is the exception, not the rule. Under a protocol frozen before any value was queried, we assembled a rule-based roster of ten countries that raised their minimum marriage age (law dates from Batyra and Pesando, 2021, and from the studies cited above), stacked DHS cohort indicators through the program's public API, and measured the bound margin over the decade before each reform. In none of the seven computable countries was that margin collapsing: the share of women married or in union by 18 changed by between $+1.6$ and -19.9 percent over the pre-reform decade, from levels of 38–53 percent in all but one country. The same data show where Brazil differs: among women aged 15–19 in union, the share declaring themselves *married*—a DHS category that includes religious and customary marriage, not civil registration—runs from 73 to 100 percent across the African and Asian reformers, against 7.5 percent in Honduras, the one Latin American consensual-union regime in the roster; Brazil's own PNADC and SINASC figures place it firmly in the latter regime. Read descriptively, this pattern is consistent with a regime account rather than a laws-do-not-work account: the same legal instrument meets formal margins of different sizes. Read jointly with Mexico, the evidence is consistent with larger registration responses where the formal margin is still quantitatively important [Bellés-Obrero and Lombardi, 2023]. It is not a general causal gradient; countries, ages, legal rules, and outcomes differ.

The gap between the locked -1.1% and no-trend -37.8% Registry estimates shows why secular declines cannot be assigned to a reform without a validated counterfactual. The grouped SINASC exercise shows the mirror image: a poorly fitting trend can produce a positive "effect." The daily design then shows that better age measurement is necessary but not sufficient. With the public data now available, the defensible endpoint is identified uncertainty. A causal next step requires exact celebration and birth dates and residence-compatible Civil Registry records, a compatible population at risk, and the same density, placebo, stability, and power gates applied before an effect is read.

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A SINASC status-at-birth exhibits

A.1 Exact-age design

Table 3: Exact-age SINASC difference in discontinuities and frozen validation gates.

Panel A. Primary and delay-sensitive estimands (percentage points)

Estimand	Estimate	SE	95% CI	p	Classification
Immediate jump change (τ)	+0.342	0.246	[-0.140, +0.824]	0.164	INCONCLUSIVE
Delay-sensitive 90-day profile	+0.323	0.327	[-0.318, +0.964]	0.328	INCONCLUSIVE

Panel B. Counterfactual diagnostics (percentage points)

Diagnostic	Cutoff age	Estimate	90% CI	p	Equivalent
Pre-law temporal placebo	16	-0.280	[-0.725, +0.165]	0.301	no
Placebo birthday	15	-0.027	[-0.453, +0.399]	1.000	no
Placebo birthday	17	+0.054	[-0.569, +0.676]	1.000	no
Placebo birthday	19	-0.064	[-0.799, +0.671]	1.000	no

Panel C. Mechanical gate verdicts

Gate	Object	Verdict
G0	Data integrity and power	PASS
G1	Density, missingness, and composition	PASS
G2	Counterfactual stability	QUALIFIED
G3	Informativeness	INCONCLUSIVE

Notes: All rows derive from observed official SINASC records. The primary sample contains 126,138 singleton live births and 1,342 recorded-married outcomes within 90 days of the mother's sixteenth birthday. τ is the 2022–2024 minus 2016–2018 change in the right-minus-left cutoff jump. The model uses triangular weights, birth-year and child-calendar-month fixed effects, and two-way clustering by municipality and exact birth date. The primary p is unadjusted; the delay-sensitive p is Holm-adjusted within the two-estimand auxiliary family. Placebo-cutoff p-values are Holm-adjusted within ages 15, 17, and 19; the temporal-placebo p is unadjusted. Equivalence requires the 90% interval to lie inside ± 0.25 points. All 13 frozen sensitivity estimates are identified and span [+0.148, +0.356] points. Qualified G2 and inconclusive G3 prohibit an unconditional causal interpretation.

A.2 Grouped-age design

Table 4: SINASC status-at-birth contrasts, mothers 10–15 versus 17–19, 2019M4–M12 (extension lock v1.0.0).

Outcome	Trended effect (95% CI)	p (Holm)	No-trend effect	MDE_{80}	TOST p	Treated events
Married (S1)	29.3% [11.3, 50.1]	0.002	−23.9%	23.8%	0.892	270
União estável (S2)	3.7% [0.9, 6.7]	0.010	−3.7%	4.1%	< 0.001	7,848
Any declared union (S3)	5.1% [2.2, 8.0]	0.001	−3.7%	4.0%	< 0.001	8,118

Notes: PPML on region-of-residence × group × month cells; the offset is births with valid status. Inference clusters by period. Column 2 uses group-specific linear trends; column 4 omits them. The TOST equivalence interval for the rate ratio is [0.85, 1.176]. Frozen diagnostics reject a causal reading of all three contrasts (Table 5); the table documents specification dependence.

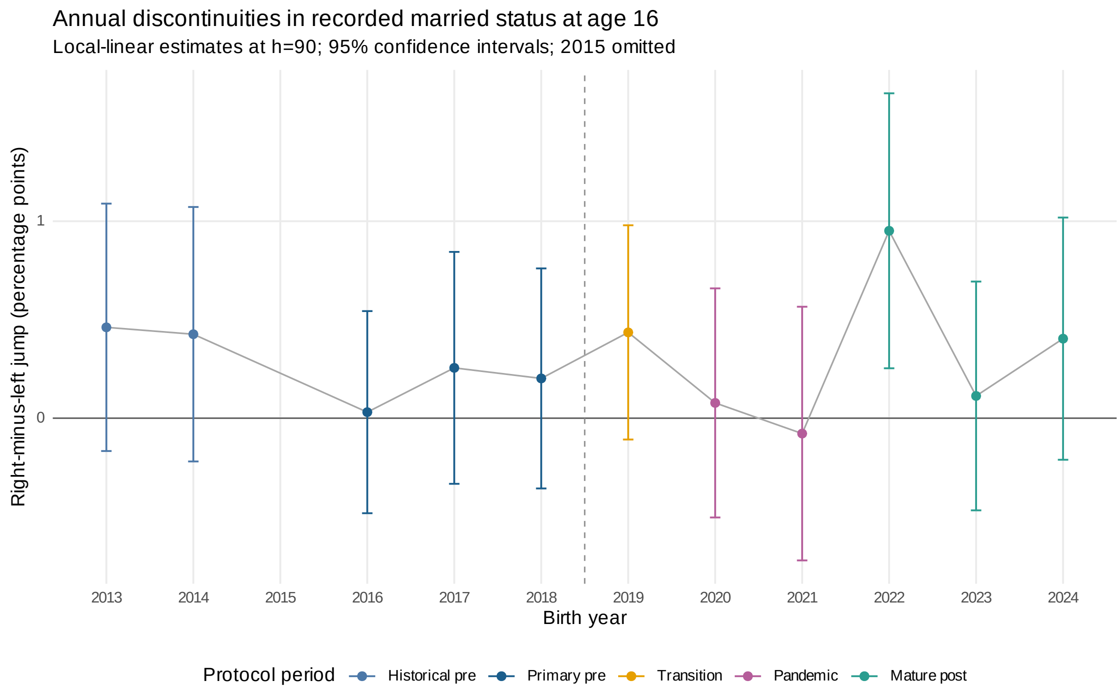

Figure 6: Year-specific age-16 discontinuities in recorded married status. Points are right-minus-left local-linear jumps within 90 days; bars are 95% confidence intervals. The vertical line marks the 2019 reform, while colors distinguish historical pre-law, primary pre-law, transition, pandemic, and mature post-law years. The incomplete 2015 file is omitted. These annual jumps are stability diagnostics, not policy effects. Source: official SINASC microdata and replication package.

Table 5: SINASC frozen diagnostics: placebo reform dates, lead test, aggregate HAC, misreporting, and fertility-composition contrasts.

Object	% effect	95% CI	p
Placebo ban April 2016	-5.9	[-21.1; 12.2]	0.497
Placebo ban April 2017	19.7	[5.9; 35.3]	0.004
Placebo ban April 2018	36.8	[16.6; 60.6]	<0.001
Brazil monthly gap, HAC(12)	24.8	[3.5; 50.4]	0.024
Unknown-status share (misreporting)	3.4	[-8.1; 16.3]	0.577
Fertility, age 15 (S4)	1.6	[-1.3; 4.6]	0.284
Fertility, age 14 (S4 rob.)	-3.0	[-6.1; 0.2]	0.137
Joint test, 24 monthly leads		$p < 0.001$	

Placebo bans use pre-2019 data only, with the true model and windows transposed to each placebo year. Fertility rows: PPML per 100,000 female residents of the same age (PNADC denominators), quarterly, Holm within family.

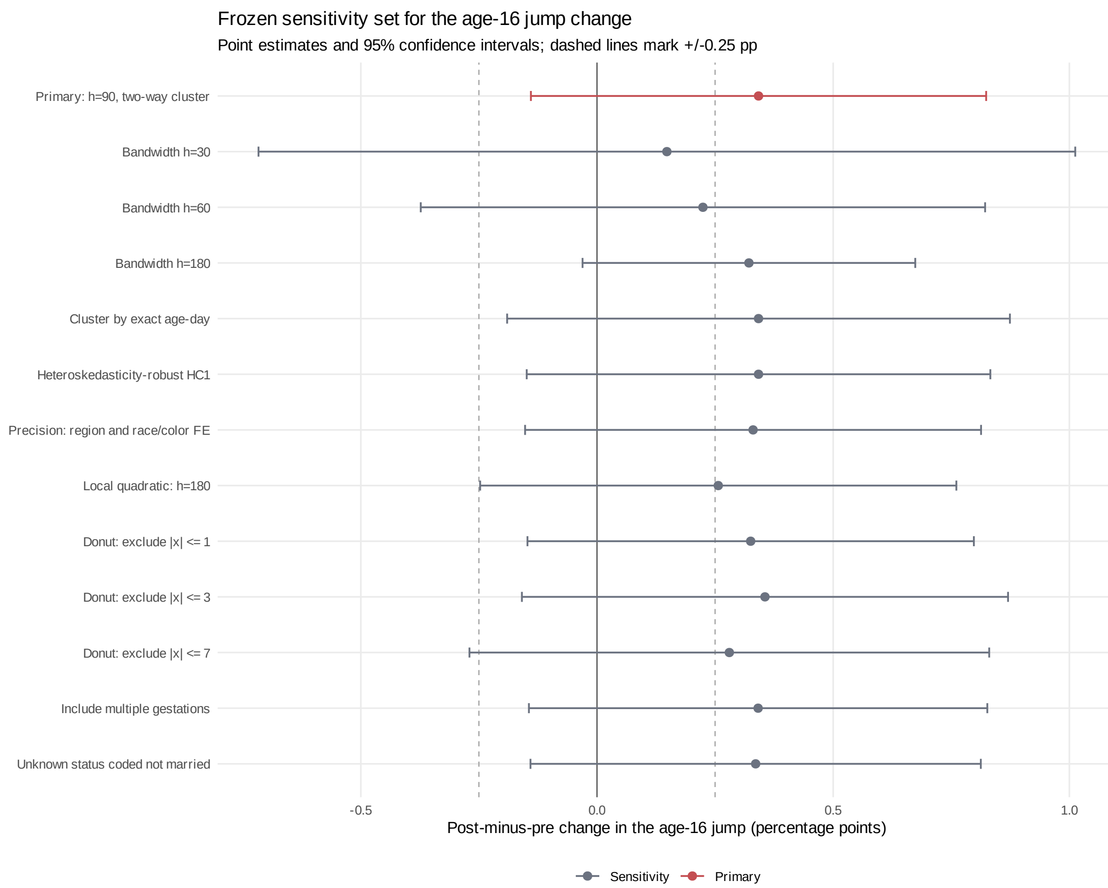

Figure 7: Frozen sensitivity set for the post-minus-pre change in the age-16 married- status jump. Every point uses observed SINASC records; specifications vary bandwidth, covariance estimator, fixed effects, polynomial order, donut window, or sample rule as fixed before estimation. Bars are 95% confidence intervals. Sensitivity estimates cannot override the qualified counterfactual gate or the inconclusive primary result. Source: replication package.

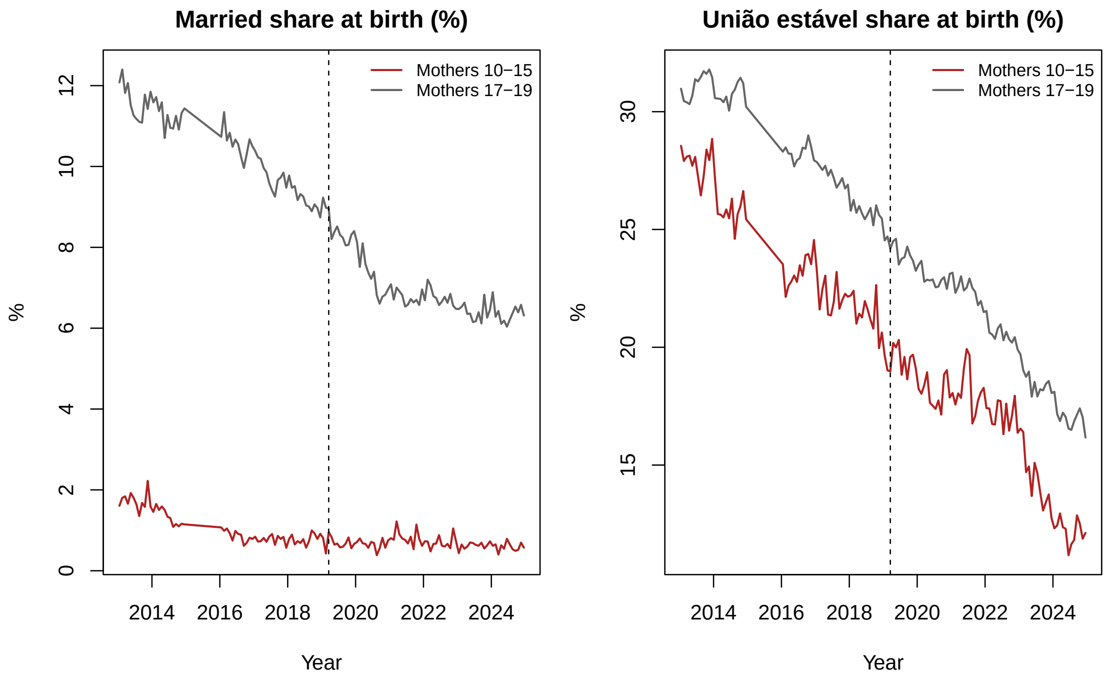

Figure 8: Declared conjugal status at birth, Brazil, monthly 2013–2024: married and *união estável* shares among mothers aged 10–15 and 17–19. Vertical line: Law 13,811 (March 2019). Source: replication package.

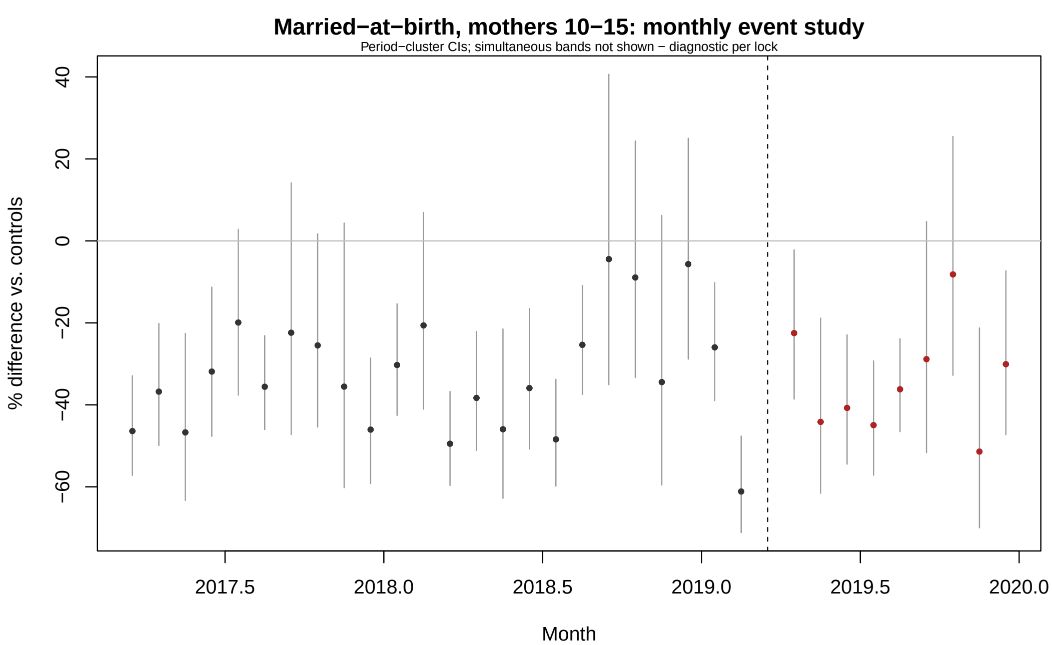

Figure 9: Married-at-birth among mothers 10–15 versus 17–19: monthly event study relative to the pooled pre-2017 baseline. Period-cluster confidence intervals; diagnostic per the grouped-age extension lock. Source: replication package.

B Registry and PNADC exhibits

Table 6: Post-result rolling-origin sensitivity to the counterfactual trend.

Model	Pre-window RMSE	Tier	Target change [95% interval]	p
Global linear	0.131	fail	+2.0 [-17.6, +23.4]	0.853
Local linear (12 quarters)	0.156	fail	+1.1 [-14.4, +18.8]	0.888
Local linear (16 quarters)	0.157	fail	+10.1 [-10.8, +37.0]	0.420
Global quadratic	0.243	fail	+12.8 [-13.6, +48.3]	0.409
Seasonal level	0.391	fail	-33.7 [-54.7, -0.3]	0.036

Notes: The target is the mean actual-minus-forecast national log-rate gap in 2019Q2–Q4, transformed to percent. RMSE uses six frozen three-quarter rolling windows before 2019. All candidates fail the frozen qualified tier. Intervals use 1,999 four-quarter moving-block bootstrap draws and are model-specific forecast intervals. This post-result exercise does not produce a causal bound.

Table 7: Trend and control-group robustness.

Specification	Controls	$\widehat{\beta}$ (SE)	p	Rate ratio	Change
Primary: age trends	17–19	−0.011 (0.077)	0.888	0.989	−1.1%
No age trends	17–19	−0.476 (0.073)	< 0.001	0.622	−37.8%
Age trends, alternative controls	18–19	−0.003 (0.079)	0.971	0.997	−0.3%
Age trends, potentially contaminated controls	16–17	−0.033 (0.070)	0.637	0.967	−3.3%

Notes: Period-cluster standard errors. The coefficient is on the log-rate-ratio scale; "Change" is $100[\exp(\widehat{\beta}) - 1]$.

Table 8: Age-specific event effects and aggregate recapture.

Age	Controls	Change	Estimated event effect (95% CI)	Aggregate recapture
15	17–19	−1.1%	−2.1 [−48.3, 31.8]	174.8
16	18–19	0.6%	40.6 [−457.7, 518.1]	—
17	18–19	3.3%	326.9 [−135.3, 781.0]	—

Notes: Event effects are fitted counts with treatment minus fitted counts without treatment, so a negative age-15 value corresponds to estimated events averted. The aggregate recapture ratio is unstable, does not track people, and is undefined in half of the bootstrap draws; it receives no causal weight.

Table 9: Primary design-based estimate for conservative coresident union.

Effect (p.p.)	95% CI	p	MDE_{80} (p.p.)	TOST p	Unweighted N (cases)
0.399	[−0.003, 0.801]	0.052	0.575	0.311	1,058,334 (57,141)

Notes: Combined standard error is 0.205 percentage points, obtained by quadrature of temporal and marginal design variance. TOST margin is ±0.5 p.p.; equivalence at 5% is not established.

Table 10: Inference triangulation for the national reform.

Method	Scale	Estimate (SE)	95% CI	p	Caveat
PPML, period clusters (primary)	Log rate ratio	−0.011 (0.077)	[−0.161, 0.139]	0.888	Only three treated post-period clusters.
PPML, region and period clusters	Log rate ratio	−0.011 (0.0004)	[−0.0115, −0.0101]	< 0.001	Five region clusters; covariance required a positive-semidefinite adjustment. Not conclusion-supporting.
PPML, region clusters	Log rate ratio	−0.011 (0.089)	[−0.186, 0.164]	0.904	Five clusters; diagnostic only.
Brazil age-15/control gap, Newey–West(4)	Aggregate log-rate gap	0.020 (0.073)	[−0.123, 0.162]	0.785	National-series approximation.
Pre-period linear-seasonal forecast, moving-block bootstrap	Forecast log-rate gap	0.020 (0.103)	[−0.185, 0.215]	0.857	Pre-period residual block bootstrap.
Prespecified-date randomization inference	Studentized PPML statistic	−0.141	—	1.000	Only four locked pseudo-dates; p-values are coarse.

Notes: Estimand scales differ as labeled. Diagnostics with inadequate cluster support are reported but receive no conclusion-supporting weight.

Table 11: Minimum detectable effects and power diagnostics.

Outcome	Target	Approx. power	SE	MDE_{80}	Caveat
Formal marriage, age 15	−10%	0.280	0.077	−19.3%	Normal approximation; period-cluster SE and three treated post quarters.
Formal marriage, age 15	−20%	0.830	0.077	−19.3%	Same calculation.
Formal marriage, age 15	−30%	0.997	0.077	−19.3%	Same calculation.
Formal marriage, age 15	−40%	1.000	0.077	−19.3%	Same calculation.
Conservative union, age 15	—	0.800	0.0021	0.575 p.p.	Temporal plus marginal design variance.

Notes: Two-sided $\alpha = 0.05$. Registry MDE is expressed as a decline; the union MDE is in percentage points.